September 28–October 2, 2026

Feedback in Practice

Connecting evidence, feedback, and shared planning to move learning forward

NEW BEDFORD HIGH SCHOOL · SCIENCE DEPARTMENT



Science teachers examine student evidence, draft proposed goals, and calibrate the next instructional cycle.

THIS WEEK'S PLC ARC

From Evidence to Proposed Goals

Our September PLC sequence closed the loop from planning and student evidence to feedback, revision, calibration, and forward planning. Course-alike teams named the learning patterns they noticed, considered which routines helped make student thinking visible, and selected a focused priority to carry into the next lesson.

> **Protected focus:** use available evidence—not a generic activity or invented number—to identify one student-learning need and the educator practice most likely to support it.

THE EVIDENCE-TO-ACTION CYCLE TEAMS CARRIED FORWARD

01 · PLAN
Plan to Notice
Identify the key thinking and evidence that will matter.

02 · REVIEW
Study Evidence
Name the learning pattern supported by student work.

03 · RESPOND
Plan Feedback
Choose a clear next action students can use.

04 · REVISE
Replan
Make one visible change to the next learning experience.

05 · CALIBRATE
Carry Forward
Select a useful routine and shared learning priority.

SEPTEMBER 28 · TEAM SYNTHESIS

Share, Calibrate & Set the Next Cycle

Teams used the full cycle to decide what to retain and revisit together.

- Share the evidence, learning pattern, feedback decision, and lesson revision.
- Calibrate a planning, feedback, or facilitation practice.
- Choose one upcoming lesson and one evidence look-for.

SEPTEMBER 29 · GOAL SETTING

Evidence Becomes a Proposed Goal

Teachers drafted one Student Learning SMART Goal and one connected Professional Practice SMART Goal.

- Begin with a BOY benchmark, assessment, student work, exit response, or observation notes.
- Use a course-alike SMART check to strengthen the connection between student need and educator action.

PLC commitment: identify the evidence, name the learning priority, select one useful practice, and return to student work to see whether the next step helped.

EVIDENCE SOURCE
Save the baseline or student work that made the learning pattern visible.

CONNECTED GOALS
Keep the Student Learning and Professional Practice SMART Goals together.

REVIEW POINT
Name the future evidence and date the team will use to check progress.

September 28–October 2, 2026

Classroom Spotlight

Students use data, discussion, and written explanations to make sense of population change

GRADE 9 BIOLOGY · SEPTEMBER 30

Reading a Population Story Through Data

In Mr. Tyler Gilbert's Biology class, students explored grey seal population data to investigate limiting factors, carrying capacity, and change over time. Graph evidence became the starting point for students' questions, calculations, and scientific explanations.

Science practices on display

NOTICE PATTERNS
Students used graphs and historical data to identify changes in a population over time.

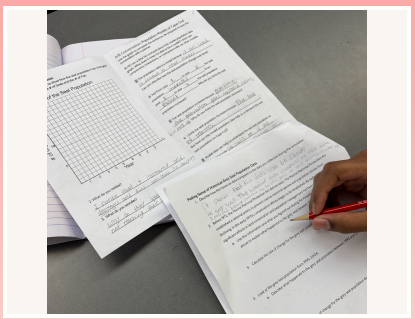
USE EVIDENCE
Calculations and data points became evidence for claims about population change.

REVISE THINKING
Written work gave students a place to clarify and strengthen their explanations.

WHAT STUDENTS WERE DOING

NOTICE · CALCULATE · EXPLAIN

- Compare historical data with data collected during a class simulation.
- Identify patterns of population increase, decrease, and stabilization on graphs.
- Use coordinates to calculate average rate of change.
- Connect a limiting factor to a data-based explanation of population size.



Student work captures graph evidence, written reasoning, and revision in progress.

STUDENTS MAKING SENSE

Data, Discussion & Scientific Reasoning

Students used grey seal population charts and related materials to make sense of a changing system. Their work asked them to compare patterns, calculate rates of change, and explain what the data suggest about population growth, decline, and stability.

Partner talk and written responses created multiple opportunities for students to share their thinking. Students used the language of limiting factor, population, carrying capacity, increase, decrease, and stabilize as they connected graph features to scientific ideas.

Learning that is visible

The student work shows a meaningful science routine: study the data, make a claim, and use evidence to explain the claim. Graphs, calculations, notes, and written responses make the reasoning process visible and give students authentic ways to communicate what they are learning.

Science habit on display: use evidence to explain how and why a population changes over time. This work brings together quantitative reasoning, ecological thinking, and clear scientific communication.

FROM GRAPH TO EXPLANATION

NOTICE	QUANTIFY	EXPLAIN
What pattern do you see in the population data?	What does the rate of change show?	Which evidence supports a claim about a limiting factor?

Students can return to the graph, calculation, and written response as they refine their explanations.

When students explain a changing system with data, they practice the core scientific habit of connecting evidence to a claim.

September 28–October 2, 2026

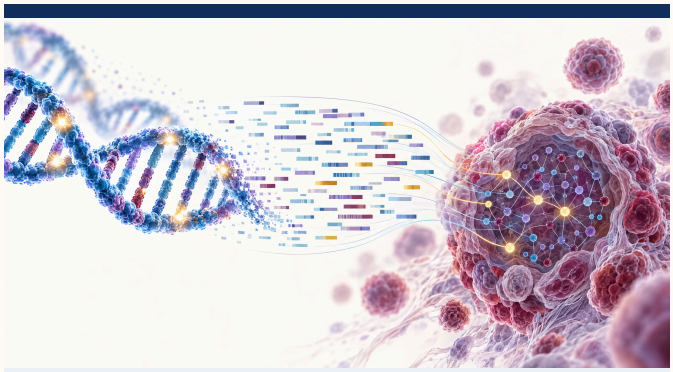
Science Corner

A closer look at cancer sequencing, signal, and scientific uncertainty

GENOMICS & CANCER DIAGNOSTICS

Sequencing Can Read a Tumor's Clues—Not Write Its Entire Story

Modern sequencing is making cancer diagnostics more precise, while also revealing why interpretation requires evidence, context, and caution.



Editorial illustration: genomic data can reveal patterns, but clinical interpretation asks which changes matter.

What sequencing can do

Next-generation sequencing (NGS) can examine many genes at once in a tumor sample, identify changes that may be relevant to diagnosis or treatment, and help clinicians monitor disease over time. It can also support testing for inherited cancer syndromes and the study of tumor DNA found in blood, often called a liquid biopsy [1,4].

The advance is not simply "reading more DNA." It is connecting sequence data with pathology, clinical history, and research evidence to ask whether a particular alteration is informative, actionable, inherited, or only part of the background variation in a tumor.

Why a mutation is not automatically a cause

Tumors can contain many DNA changes. Some **driver mutations** contribute to cancer growth; many **passenger mutations** accumulated as cells divided but do not themselves drive the disease. In a major whole-genome study across 38 tumor types, cancer genomes averaged about four to five identified driver mutations when coding and noncoding elements were combined—yet drivers were not identified in roughly 5% of cases [2].

> ***A sequencing result is therefore a scientific lead, not a stand-alone verdict. Researchers compare the variant with known cancer databases, the tumor type, the patient's clinical picture, and—when possible—functional evidence.***

AN INTERPRETATION CHECK

Three Questions After Sequencing

1. What change was found?
2. Could it affect a cancer-related gene or pathway?
3. Does the tumor, clinical, and research context support meaning?

Moving from DNA data to a diagnosis is an evidence check—not a single-mutation match.

Why environment belongs in the conversation

Cancer-related genetic changes can arise through random errors as cells divide, inherited susceptibility, or environmental exposures such as tobacco smoke, ultraviolet radiation, some chemicals, and certain viruses [4]. Sequencing can detect *mutational signatures*—recurring patterns that reflect DNA damage or repair processes. Some signatures are associated with external or internal processes, but many still have unknown causes [3].

Two questions, two kinds of testing

Sequencing a tumor to find biomarkers is different from inherited cancer-risk testing. Tumor profiling asks about changes in cancer cells that may help guide treatment; inherited testing looks for changes present throughout the body that may affect risk for a person and biological relatives. A tumor result can sometimes lead to follow-up testing or genetic counseling, but it does not alone prove that a variant was inherited [4].

Precision means working with complexity

Different cells within the same tumor may carry different changes, and genetic profiles can evolve. That tumor heterogeneity, together with limits in sample quality, data interpretation, cost, privacy, and access, helps explain why a genomic test may guide a next question rather than provide a single final answer [1,2].

Scientists and clinicians therefore ask whether a variant is known to affect a cancer-related gene or pathway, whether it fits the tumor type, how consistently it is observed, and whether research evidence supports its interpretation [1,5]. For students, this is a powerful reminder: better tools sharpen questions, but responsible conclusions depend on comparison, evidence, and an understanding of what the data cannot yet tell us.

REFERENCES

[1] Ghoreyshi, N., Heidari, R., Farhadi, A., et al. (2025). *Next-generation sequencing in cancer diagnosis and treatment: clinical applications and future directions. Discover Oncology, 16*, 578.
[2] ICGC/TCGA Pan-Cancer Analysis of Whole Genomes Consortium. (2020). *Pan-cancer analysis of whole genomes. Nature, 578*, 82–93.
[3] Alexandrov, L. B., Kim, J., Haradhvala, N. J., et al. (2020). *The repertoire of mutational signatures in human cancer. Nature, 578*, 94–101.
[4] National Cancer Institute. (2024). *The Genetics of Cancer.*
[5] Raphael, B. J., Dobson, J. R., Oesper, L., & Vandin, F. (2014). *Identifying driver mutations in sequenced cancer genomes. Genome Medicine, 6*, 5.

September 28–October 2, 2026

Science Team Gratitude

Celebrating the generosity, collaboration, and care that strengthen our department

THANK
YOU
SCIENCE
TEAM

A DEPARTMENT THAT WORKS AS A TEAM

Every Shared Contribution Moves Learning Forward

Our PLCs, professional learning, classroom experiences, and daily operations are stronger because colleagues share their expertise, time, materials, and feedback. This page celebrates just a few of the many people who make New Bedford High School Science a supportive professional home.

01 OpenSciEd Storyline Leadership

Thank you, Ms. Grace Stephenson, for leading colleagues as they implement the Grey Seal population activity. **Thank you, Ms. Seannine Tveit,** for sharing the aligned data assignment that helps students engage with the OpenSciEd storyline.

02 Supplies Ready for Learning

Thank you to Mr. Tyler Gilbert, Mr. Nathaniel Walker, Ms. Cassidy Barrett, and all colleagues who proactively helped distribute supplies and materials so science classrooms are ready for learning.

03 Welcoming PLC & PD Spaces

Thank you to Ms. Cassidy Barrett, Mr. Tyler Gilbert, and Ms. Patricia Gifford for welcoming the science team into their classrooms for PLCs and professional development. We appreciate your generosity—and will work together to keep these spaces tidy.

04 Classroom Homes for Floating SPED Staff

Thank you to science team members including Mr. Kevin Flynn and Mr. Jonathan Way for welcoming floating special education colleagues into their classrooms and helping create consistent, supportive home bases.

05 Feedback That Helps Us Grow

Thank you to every colleague who gives feedback to strengthen our PLC, PD, and collaborative learning experience. Your feedback is welcomed wholeheartedly and helps us improve together.

The signature of the New Bedford High School Science Department is a team that supports one another, shares generously, and grows together.

"The willingness to help, share, host, prepare, and offer honest feedback is what makes our department a team and a family."

WITH APPRECIATION FOR EVERY MEMBER OF THE NEW BEDFORD HIGH SCHOOL SCIENCE TEAM

Wednesday, September 30, 2026

Feedback That Moves Learning

Student ownership, useful feedback, and a current ASPEN gradebook

THANK YOU

SCIENCE TEAM

for your attentive participation and great attendance during our professional development session.

SEPTEMBER 30 PROFESSIONAL DEVELOPMENT

One Shared Language for Feedback

Our PD explored how feedback and an up-to-date ASPEN gradebook can tell students the same story: where they are, what is working, and what they need to do next. The goal is not more comments—it is more useful information that moves learning forward.

WED
9·30
Feedback & ASPEN

THE FOUR QUALITIES OF FEEDBACK THAT MOVES LEARNING

01
Timely
Close enough to the learning that students can still use it.

02
Specific
Names what is working or what needs attention.

03
Actionable
Gives students a clear next step they can take.

04
Connected
Links to the learning goal, criteria, rubric, skill, or standard.

TIMELY GRADING IS TIMELY FEEDBACK

Keep ASPEN Current for Students & Families

Please enter grades for student assignments into ASPEN in a timely manner. A current gradebook helps students and families understand what has been completed, what is missing, where revision or retake work has been reflected, and what needs attention next.

CLASSROOM FEEDBACK → STUDENT ACTION → CURRENT ASPEN EVIDENCE → NEXT DECISION

SPECIAL GRATITUDE

Thank You, Mr. Sean McNulty

Thank you for taking time to visit our PLCs and help colleagues set up the standardized ASPEN special grading codes highlighted during the PD.

Abs	**Absent**	Not exempt from calculation
E	**Excused**	Exempt from calculation
M	**Missing Grade**	Not exempt from calculation

MAKE TITLES CLEAR
Use assignment titles students and families can understand.

ENTER OR UPDATE
Record assignment grades and reflect revisions or retakes when appropriate.

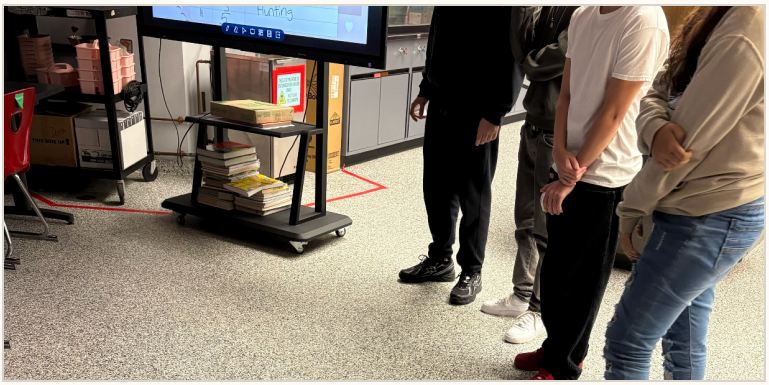
REVIEW THE STORY
Look at ASPEN through the eyes of a student or family member.

When classroom feedback and ASPEN are both current, students have a clearer path from evidence to action.

September 28–October 2, 2026

Classroom Spotlight

Environmental Science students model how limiting factors shape a population



Students participate in the bear-population simulation as the class tracks resources and population change.

ENVIRONMENTAL SCIENCE · MS. CASSIDY BARRETT

“A Bear-y Good Habitat”

Students transformed the classroom into a bear-population simulation to investigate how limiting factors affect survival, population size, and carrying capacity. Each round made an ecological relationship visible: a bear needed to find a resource in order to survive and reproduce.

Learning goal: model how resource availability affects a population; identify changes over time; and use data to explain carrying capacity.

THE SHARED MODEL

Food	Water	Shelter	Population
Energy for survival	A changing resource	Protection and habitat	The shared data story

LEARNING THROUGH A MODEL

From Roles to Population Data

Students took on the roles of bears and resources, then followed a consistent routine for each simulated year: choose a role, reveal at the class signal, match one bear to one resource, and record the outcome. The group’s experience became a class data set that students could analyze together.

Changes in resource availability and environmental events, including drought and predation, gave students a concrete way to see why a population may increase, decrease, or stabilize. The simulation made the relationship among resources, limiting factors, and survival something students could experience before putting it into scientific language.

> ***“How do resources and limiting factors affect population size and carrying capacity?”***

SCIENCE PRACTICES IN ACTION

Notice · Record · Graph · Explain

NOTICE	Compare what happened to the bear population from one simulated year to the next.
RECORD	Use shared class data to document population change and the event connected to each round.
GRAPH	Translate the model into a visual record of increases, decreases, and patterns over time.
EXPLAIN	Use evidence from the data to connect resource availability to carrying capacity.

By moving from an active simulation to graphs and evidence-based explanations, students experience ecology as a system of connected causes and effects.